SAMARTH BIRAJDAR

Computer Science Engineer | B.Tech — 3rd Year

+91 9922508908 samarth.birajdar31@gmail.com github.com/samarth3107

SKILLS

Languages
C • Python • HTML • MySQL

CS Concepts
Data Structures • Algorithms • DBMS • OOP • Machine Learning • NLP

ML / AI
Random Forest • SVM • XGBoost • TF-IDF • PSO • DE • GA • WOA • GWO

Tools & Tech
RPA (AutomationEdge) • Git / GitHub • IoT (Basic)

Soft Skills
Research • Analytical Thinking • Problem Solving • Team Collaboration

Languages Known
English • Marathi • Hindi

CERTIFICATIONS

Introduction to Databases
Meta / Coursera • Feb 2025
coursera.org/verify/1GHPYC011PEJ

AutomationEdge Foundation (RPA)
AutomationEdge • Sep 2025
Cert No: 20756C3R6BOI1Z4

Coding and Programming
Samsung Innovation Campus • Nov 2025 – Mar 2026
Telecom Sector Skill Council

CONTACT

PROFESSIONAL SUMMARY

Motivated Computer Science undergraduate (3rd Year, CGPA 7.11) at Pimpri Chinchwad University with hands-on research experience in Machine Learning, NLP, and Bio-Inspired Optimization. Co-authored a research paper on News Category Classification using metaheuristic algorithms achieving 87% accuracy with DE-SVM. Built an ML-based Hospital Mortality Prediction System. Holds 3 industry certifications from Meta, AutomationEdge, and Samsung. Seeking a challenging role in software development, ML, or data engineering.

EDUCATION

2023 – Present PCU, Pune	**B.Tech – Computer Science | Pimpri Chinchwad University**	**CGPA: 7.11**
2022 – 2023 Maharashtra Board	**HSC – Mahatma Gandhi Global Village, Boramni**	**74.4%**
2020 – 2021 Maharashtra Board	**SSC – St. Joseph High School, Solapur**	**71%**

PROJECTS

Hospital Mortality Rate Prediction System | *Python, ML, EHR Data*
Guide: Dr. Sachin Jadhav | Pimpri Chinchwad University

- Built an ML system to predict in-hospital patient mortality using EHR data including vital signs, lab results, comorbidity indexes, and patient demographics.
- Implemented Random Forest, SVM, and Neural Networks with a Decision Fusion layer to improve overall prediction accuracy and resilience.
- Applied data preprocessing pipeline: missing value handling, normalization, feature engineering, and cross-validation with hyperparameter tuning.
- Evaluated using Accuracy, Precision, Recall, F1-Score, and AUC-ROC in a 3-member team; managed feature selection pipeline and model training.

Student Record Management System | *HTML, MySQL*

- Developed a web-based CRUD application to store, retrieve, and manage

Phone
+91 9922508908
Email
samarth.birajdar31@gmail.com
GitHub
github.com/samarth3107

student academic records using MySQL and HTML forms.
- Implemented search and filter functionality to query records by roll number, class, and result status.

RPA Automation Workflow | *AutomationEdge RPA*
- Designed and deployed an RPA workflow during AutomationEdge Foundation training to automate a repetitive data-entry task.
- Reduced simulated manual processing time by automating form filling and data transfer steps between applications.

ACADEMIC RESEARCH

News Category Classification using Bio-Inspired Optimization

Authors: *Prathamesh Jangam, Samarth Birajdar, Sanskar Kumbhare*

Pimpri Chinchwad University, 2025 | Domain: ML, NLP, Metaheuristic Optimization | Status: Manuscript in Preparation

Benchmarked 6 ML classifiers (RF, SVM, XGBoost, LR, DT, NB) optimized with 5 bio-inspired algorithms (PSO, DE, GA, WOA, CGWO) on the UCI Spambase dataset (4,601 instances, 57 features).

Best results: DE-SVM achieved 87% accuracy; WOA-Random Forest achieved 85%. Validated with Friedman and Nemenyi statistical tests.

Key finding: Metaheuristic optimization outperforms grid/random search, reducing manual tuning effort while improving generalization.

CAREER OBJECTIVE

Seeking a long-term role in software development, machine learning, or data engineering — leveraging my research background in NLP and bio-inspired optimization, along with practical skills in Python, MySQL, and RPA — within a technology-driven organization where I can contribute meaningfully and grow as a CS professional.